22 Jan. 2023 Microprocessor Systems - Final /fall 2023

Student ID: Answers Sheet

Student Name:

رقم الشعبة:

Part 1: MULTIPLE CHOICES QUESTIONS (33 marks)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22

QUESTION #2(5 pointss)

A. Write the equivalent 8086 assembly code for :

if(x > y) S=x+y else Y++;

Assume X, Y ,S signed byte memory locations

B. Write a procedure to find number of ones in register AL and return the result value into register DL.

Question #3

Using BIOS interrupt, and DOS interrupts write assembler codes to print on the screen "Albaqa' Applied University" in position (15, 40).

Part 1: MULTIPLE CHOICES QUESTIONS (33 marks)

1. The interrupt vector contains two words (pointers) which specify the address of:

   A. Interrupt vector table B. Interrupt service routine C. Interrupt type D. I/O ports

2. The physical address of interrupt vector associated with INT 8 is:

   A 00016H B . 00020H C. 00040h D. 00015H

3. The size of interrupt vectors table is ------

   A. 1024 bytes B.4 bytes C. 64 bytes D. 128 bytes

4. The binary value of AL and carry flag (CF)after the following instructions have executed is ....

```
MOV AL,57H
SHL  AL,1
```

   A. AL=1011110; CF=1 B. AL=10101110 ;CF=0
   C. AL=00101011;CF=1 D.. AL=00101011;CF=0

5. The binary value of AL and CF after the following instructions have executed is ....

```
MOV CL,2
STC
MOV AL, 5BH
RCR AL,CL
```

   A. AL=11010110 ; CF=1 B. AL=11010110 ; CF=0
   C. AL=11010110 ; CF=1 D. AL=00101011;CF=0

6. In the following instruction sequence, show the value of AL, in binary:

```
MOV AL,3FH
XOR AL,FFH
```

   A. AL= 11000000 B. AL= 11010000 C. AL= 00001111 D. AL= 11110000

7. If there is a carry from lowest 4 bit during addition, ______ flag sets.

   A. Auxiliary carry B. Carry C. Over flow D. Sign

8. The result of division instruction DIV BL is stored in register pair

   A. AH:AL B. AX:DX C. CX:AX D. AX:CX

9. Find the content of register AL after execution of the following code:

```
MOV  AL,0FH
SHR  AL,1
ADC  AL,1
```

   A. 9 B. 8 C. 7 D. 10

10. Find the content of memory location X after execution of the following code

```
X DB   0FH
MOV  AX,11
MOV  BL,3
DIV  BL
OR  AH,30H
AND  X,AH
```

   A. X=2 B. X=0 C. X= '0' D. X='2'

11. Find the content of AL registers after execution of the following code:

```
MOV  AX,FF02H
MOV  BL,4
MUL  BL
XOR  AL,FFH
```

A. 5FH B. 7FH C. F7H D. 8FH

12. LOPPE instruction repeat a group of instruction while ----------------

A. (CX=0)&(ZF=0) B. (CX=0) & (ZF≠0) C. (CX≠0) &(ZF=1) D. (Cx≠0) & (ZF=0)

13. The value of CX and AL after executing the following code is:

```
    MOV AL, FEH
    MOV CX, 5
L1: ADD AL, 1
    LOOPNZ  L1
```

A. CX=3; Al=1 B. CX=3; Al=0 C. CX=3; Al=0 C. D. CX=3; AL=2

14. To compare two strings in equality we use prefix ----------before instruction CMPSB

A. REP B. REPNE C. REPE D.REPNZ

15. The equivalent instructions of STOSW; assuming DF flag =0:

A. MOV AX,[SI]; INC SI; INC SI
B. MOV AX,[SI]; DEC SI
C. MOV [DI],AX; INC DI; INC DI
D. MOV [DI],AL; DEC DI

16. **What will be appears on the screen after executing the following code:**

```
STR DB "ABCDEF", 0DH, "AB"
    DB 'XYZ' , '$'
MOV AH, 9
MOV DX, OFFSET STR
INT 21H
```

A) ABXYZF B) ABCDEF C) ABCDEF D) XYZABF

17. The values of array Y and register AL after executing the following code is:

```
X DB   'ABC+CDK'
Y DB    5 DUP('#')
CLD
MOV SI, OFFSET X
MOV DI, OFFSET Y
MOV CX,3
REP MOVSB
LODSB
```

A) Y=ABC## , AL='+' B) Y=ABC+CD# ; AL='C' C) Y=ABC#*CD; AL=D D) Y=ABC#*,AL=D

18. To transfer value (45H) 8-bit data from input port addressed by 0CF5H to register AL in microprocessor; the following instructions can be used:

A. MOV DX, 0CF5H; IN AL, DX
B. MOV BX, 0CF5H; IN AL,BX
C. MOV DX, 0CF5H; MOV AL, 45H; OUT DX, AL
D. MOV BX, 03FCH; OUT DX, AL

19. What is the value of CX REGISTER after execution the following codes?

```
STR DB "AB+DE"

MOV DI, OFFSET STR
MOV CX,5
MOV AL,'+'
REPNE SCASB
```

A.1 B. 3 C. 4 D. 2

20. - During the execution near call instruction the ---------------- changed

A. IP, DS B. only CS C. IP,CS D. only IP

21. The value of Y after executing the following code is

```
X  DB  2,5,7, 4,7,8
Y  DB  ?
MOV  BX,OFFSET X
MOV CX, 2
MOV  AL,0
L:  CALL SUM
LOOP L
MOV Y, AL
HLT
SUM PROC
        ADD  AL, [BX]
        INC BX
        RET
SUM ENDP
```

A) 11 B) 08h C) 0 D) 07h

22. What is the value of SI after executing the following code if the DF=0?

```
MOV SI, 201H
LODSW
```

A. 302H B. 299H C. 200H D. 203H

Part2: Essay Questions

QUESTION #2(5points)

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QUESTION#4(7 points)

A. Explain 2 differences between Asynchronous and Synchronous serial communication

B. if the device can transmit 10 char per second find the baud rate , if Data format : 1 *start bit,7 data, 1 parity, 2 stop bits*

C. Interface 8- bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X. Assume the address of the port is 08F0H.

a) Draw block circuit diagram circuit, showing address decoding logic, and control signals

b) Write a program to read the status of switches and store the result into X memory location

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